

EXP 26 – Fruit and Its Color Using Backtracking

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Aim

To write a Prolog program that stores fruits and their colors and uses backtracking to display the color of a given fruit or list all fruits of a given color.

Algorithm

1. Define facts fruit_color(Fruit, Color) for each fruit.
2. To display the color of a fruit:
 - Match the fruit with its color using the fact.
 - Print the result using format.
3. To display fruits of a particular color:
 - Use forall to iterate through all fruits having that color.
 - Prolog backtracks automatically to find all matches.
4. Print each fruit using format.

Source Code

```
prolog

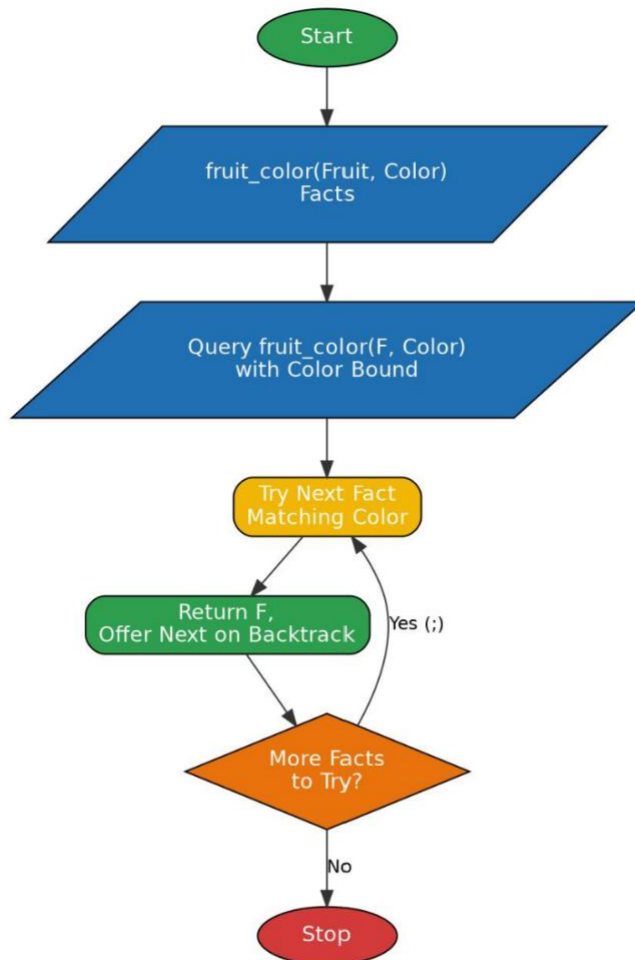
fruit_color(apple, red).
fruit_color(banana, yellow).
fruit_color(grape, green).
fruit_color(orange, orange_color).
fruit_color(mango, yellow).
fruit_color(cherry, red).

show_color(Fruit) :-
    fruit_color(Fruit, Color),
    format("~w is ~w~n", [Fruit, Color]).

show_fruits(Color) :-
```

```
forall(fruit_color(Fruit, Color),
      format("~w~n", [Fruit])).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.
?- [fruits].
true.
?- show_color(apple).
apple is red
true.
?- show_color(mango).
mango is yellow
true.
?- show_fruits(yellow).
banana
mango
true.
?- show_fruits(red).
apple
cherry
true.
```

Result

The program correctly displays the color of any fruit when queried and lists all fruits of a given color using Prolog's backtracking mechanism.